ID: dd349efc

Participants' Evaluation of the Likelihood That Robots Can Work Effectively in Different Occupations

Occupation	Somewhat or very unlikely (%)	Neutral (%)	Somewhat or very likely (%)
television news anchor	24	9	67
teacher	37	16	47
firefighter	62	9	30
surgeon	74	9	16
tour guide	10	8	82

Rows in table may not add up to 100 due to rounding.

Georgia Tech roboticists De'Aira Bryant and Ayanna Howard, along with ethicist Jason Borenstein, were interested in people's perceptions of robots' competence. They recruited participants and asked them how likely they think it is that a robot could do the work required in various occupations. Participants' evaluations varied widely depending on which occupation was being considered; for example, _____

Which choice most effectively uses data from the table to complete the example?

A. 47% of participants believe that it is somewhat or very likely that a robot could work effectively as a teacher, but 37% of respondents believe that it is somewhat or very unlikely that a robot could do so.

B. 9% of participants were neutral about whether a robot could work effectively as a television news anchor, which is the same percent of participants who were neutral when asked about a robot working as a surgeon.

C. 82% of participants believe that it is somewhat or very likely that a robot could work effectively as a tour guide, but only 16% believe that it is somewhat or very likely that a robot could work as a surgeon.

D. 62% of participants believe that it is somewhat or very unlikely that a robot could work effectively as a firefighter.

ID: 8fbed1cb

When the Vinland Map, a map of the world purported to date to the mid-1400s, surfaced in 1957, some scholars believed it demonstrated that European knowledge of the eastern coast of present-day North America predated Christopher Columbus's 1492 arrival. In 2021, a team including conservators Marie-France Lemay and Paula Zyats and materials scientist Anikó Bezur performed an extensive analysis of the map and the ink used. They found that the ink contains titanium dioxide, a compound that was first introduced in ink manufacturing in the early 1900s. Therefore, the team concluded that _____

Which choice most logically completes the text?

A. mid-1400s Europeans could not have known about the eastern coast of present-day North America.

B. the Vinland Map could not have been drawn by mid-1400s mapmakers.

C. mapmakers must have used titanium compounds in their ink in the 1400s.

D. there isn't enough information to determine when the ink was created.

ID: 55688b3c

Dutch painters in the sixteenth and seventeenth centuries often showed tables filled with large wheels of cheese or carved shards of butter. Some art historians, noting that dairy products were a major component of the Dutch diet, interpret these depictions as reflections of everyday Dutch eating habits. However, a group of researchers recently reviewed hundreds of food-related paintings and found that lemons—which could only be acquired in the Netherlands at great cost, since they had to be imported from warmer climates—feature in Dutch paintings of the period more than three times as frequently as dairy products do, thereby casting doubt on the idea that _____

Which choice most logically completes the text?

A. dairy products were a more significant component of the Dutch diet of the period than lemons were.

B. food was a more popular subject among Dutch painters than it was among painters from other countries at the time.

C. depictions of food in Dutch paintings of the period should be taken as realistic representations of Dutch eating habits.

D. Dutch painters of the period may have depicted foods for symbolic reasons rather than to show what Dutch people typically ate.

ID: d0f51067

Modern dog breeds are largely the result of 160 years of owners crossbreeding certain dogs in order to select for particular physical appearances. Owners often say that some breeds are also more likely than others to have particular personality traits—basset hounds are affectionate; boxers are easy to train—but Kathleen Morrill and colleagues found through a combination of owner surveys and DNA sequencing of 2,000 dogs that while physical traits are predictably heritable among purebred dogs, behavior varies widely among dogs of the same breed.

Which choice best states the main idea of the text?

A. Dog breeds would not exist without many years of human intervention in dogs' reproduction.

B. Research fails to confirm a commonly held belief about dog breeds and behavior.

C. The dog breeds most popular among owners have often changed over the past 160 years.

D. A study of dog breeds is notable for its usage of both opinion surveys and DNA sequencing.

ID: d1539546

Tides can deposit large quantities of dead vegetation within a salt marsh, smothering healthy plants and leaving a salt panne—a depression devoid of plants that tends to trap standing water—in the marsh's interior. Ecologist Kathryn Beheshti and colleagues found that burrowing crabs living within these pannes improve drainage by loosening the soil, leading the pannes to shrink as marsh plants move back in. At salt marsh edges, however, crab-induced soil loosening can promote marsh loss by accelerating erosion, suggesting that the burrowing action of crabs _____

Which choice most logically completes the text?

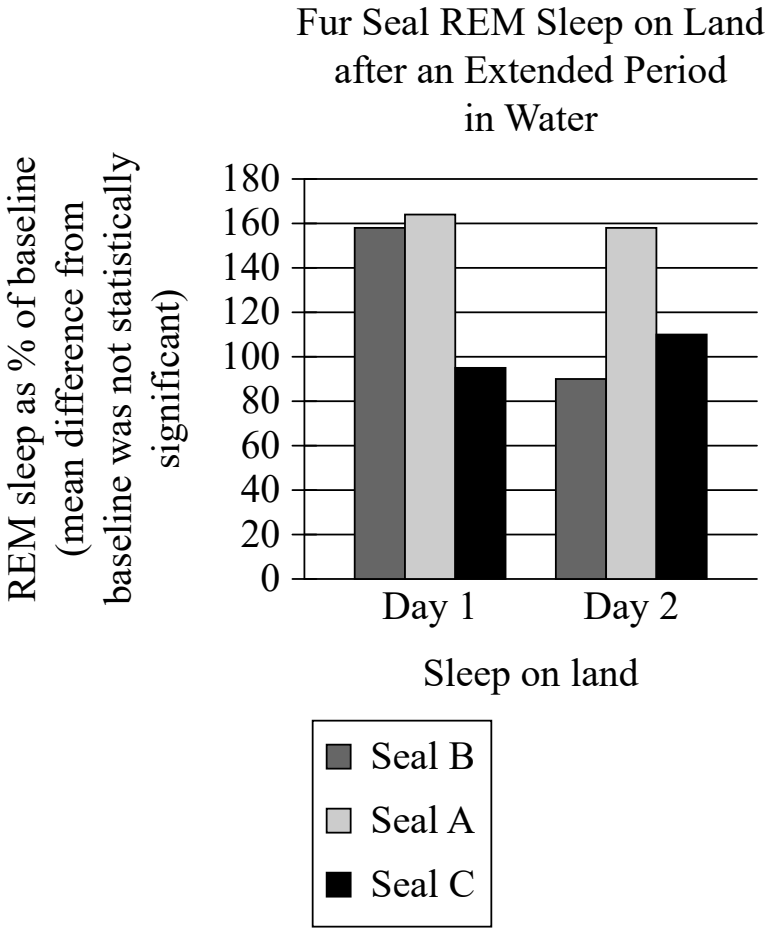
A. can be beneficial to marshes with small pannes but can be harmful to marshes with large pannes.

B. may promote increases in marsh plants or decreases in marsh plants, depending on the crabs' location.

C. tends to be more heavily concentrated in areas of marsh interiors with standing water than at marsh edges.

D. varies in intensity depending on the size of the panne relative to the size of the surrounding marsh.

ID: ccb1ab92



Research suggests that REM sleep in animals is homeostatically regulated: animals compensate for periods of REM sleep deprivation by increasing subsequent REM sleep. When on land, fur seals get enough REM sleep, but during the weeks they're in the water, they get almost none. In a study of fur seals' sleep habits, researchers recorded the REM sleep (as a percentage of baseline) of fur seals once they had returned to land. They concluded that REM sleep may not be homeostatically regulated in fur seals, citing as evidence the fact that the seals in the study _____

Which choice most effectively uses data from the graph to complete the text?

A. didn't show significantly less REM sleep during the second day after returning to land than they did during the first day.

B. showed no significant differences from one another in baseline levels of REM sleep.

C. didn't consistently demonstrate a significant increase in REM sleep after their period of deprivation in the water.

D. showed no significant difference between REM sleep after returning to land and REM sleep while in the water.

ID: 156ff681

Many governments that regularly transfer money to individuals—to provide supplemental incomes for senior citizens, for example—have long done so electronically, but other countries typically have distributed physical money and have only recently developed electronic transfer infrastructure. Researchers studied the introduction of an electronic transfer system in one such location and found that recipients of electronic transfers consumed a different array of foods than recipients of physical transfers of the same amount did. One potential explanation for this result is that individuals conceive of and allocate funds in physical money differently than they conceive of and allocate funds in electronic form.

Which finding from the study, if true, would most directly weaken the potential explanation?

A. Recipients of electronic transfers typically spent their funds at a slower rate than recipients of physical transfers did.

B. Nearly every recipient of an electronic transfer withdrew the entire amount in physical money shortly after receiving the transfer.

C. Recipients of physical transfers tended to purchase food about as frequently as recipients of electronic transfers did.

D. Some recipients of physical transfers received small amounts of money relatively frequently, while others received large amounts relatively infrequently.